LLM-Based Anomaly Detection in System Logs

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Abstract

A milestone update will be given on how Large Language Models can be used to identify problems in system and network logs. The practical application that will be used is a web application designed using the MongoDB, Express, React, and Node (MERN) stack. The framework that will be used to scan the network logs is called LangChain combined with Claude. The topic number in this syllabus is: 14. LLM-Based Anomaly Detection in System Logs. So far the application has been built with Backend Logging using the Javascript logging Library Winston which so far is able to log backend API requests and responses. LangChain framework has been implemented to some extent and is able to scan logs. However more fine-tuning needs to be done as it is not consistent and there are still bugs and issues with thee Claude API.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; *Neural networks*; • **Security and privacy** → **Intrusion/anomaly detection and malware mitigation**; • **Software and its engineering** → System administration.

Keywords

large language models, anomaly detection, log analysis, system monitoring, LangChain, natural language processing, security analysis, network logs

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1 Introduction and Problem Statement

1.1 Motivation

The motivation for this project is to explore how LLMs can be used to identify problems in system and network logs, which can be a valuable tool for system administrators and security professionals.

1.2 Significance

The significance of this project is that it can help to improve the efficiency and effectiveness of log analysis, which can lead to faster identification of issues and improved security.

1.3 Objectives

The objectives of this project are to develop a web application that can scan network logs using LangChain and Claude, and to evaluate the performance of the application in identifying anomalies in the logs.

2 Related Work

Recent advances in applying large language models to log analysis have shown promising results. Guan et al. [1] introduced LogLLM, demonstrating how LLMs can effectively detect anomalies in system logs through natural language understanding capabilities. This work provides a foundation for leveraging pre-trained language models in automated log monitoring systems.

3 Methodology

3.1 Techniques

3.2 Evaluation Metrics

4 Preliminary Results

Initial findings with brief analysis.

5 References

Proper ACM reference styles

References

[1] Wei Guan, Jian Cao, Shiyou Qian, Jianqi Gao, and Chun Ouyang. 2025. LogLLM: Log-based Anomaly Detection Using Large Language Models. arXiv:2411.08561 [cs.SE] https://arxiv.org/abs/2411.08561

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